Customer Segmentation via Data Mining Techniques: State-of-the-Art Review



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Abstract Customers are more vigilant, intelligent, and dynamic in society. They change their preferences and habits according to their needs. Knowing the needs of customers is an important part of marketing where a company should discover the loyal customers in this heterogeneity. The concept of dividing heterogeneity into homogeneous forms is termed as customer segmentation. Customer segmentation is an integral part of marketing where companies can easily develop relationships with customers with a huge set of customer data in an organized manner. Understanding the customer's hidden knowledge is a resourceful idea of computational analysis where accurate information could be optimized for the taste and preference of the customer. This type of computational analysis is termed as data mining. This paper discussed on a systematic review of customer segmentation via data mining techniques. It is a systematic review of supervised, unsupervised and other data mining techniques used in segmentation.

Keywords Customer segmentation · Data mining · Supervised · Unsupervised

1 Introduction

Understanding consumer behaviour is a resourceful idea in marketing that makes customers profitable. Always, the manufacturer provides high-quality goods or services for customers to fulfil their needs and wants by providing adequate knowledge. Basically, the needs and wants of customers are closely observed through their habits and preferences. So, knowledge is an important asset for companies to make customers loyal. Any marketer should assemble the information seamlessly to satisfy them by providing customized services at each point of delivery to avoid negative

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J. Nayak et al. (eds.), *Computational Intelligence in Data Mining*, Smart Innovation, Systems and Technologies 281, https://doi.org/10.1007/978-981-16-9447-9_38

reaction from consumers [1]. Over the years, consumers' behaviour has changed continuously. Now, consumers are more volatile than before. Often, they change their habits and preferences. Therefore, it is impossible for a seller or manufacturer to identify the consumer's needs and wants in the mass markets. The idea of dividing the market into various groups or sub-groups is typically known as segmentation. The concept of segmentation is justified and explained by different experts to identify the needs and wants of customers rationally. This strategic application of market targeting will ensure to anticipate consumer reaction, because they may have varied preferences for consuming goods or services according to their profile [2]. Nevertheless, the selection of segmentation techniques consistently depends on the variables input, such as the geographic, demographic, behavioural, or psychological profile of consumers forecasted with some statistical or non-statistical approaches.

According to Smith [3], segmentation is a distinctive marketing strategy closely associated with product differentiation and homogeneity. The customer may obtain a variety of alternatives from manufacturers. In this diversified market structure, manufacturers may get confused about selecting or retaining the customer. To attract and retain customers, often marketers adopt selective techniques through advertising or sales promotion rather than to understand the customer's motives. In the generalization of the mass market, it is difficult to identify the needs and wants of the customer through all kinds of promotional techniques. Therefore, customer segmentation could be a choice for the marketer to provide preferential goods or services to the customer. The basic idea of customer segmentation is to cluster/group customers to identify, understand and target their needs. This concept of customer segmentation was initially introduced by Smith in 1956 as an unconventional technique for product differentiation strategy. A segment or group of customers can be depicted as a set of customers who have similar types of demographic, psychological, and behavioural profiles [4]. Now the selection of segmentation techniques is a sophisticated area of research in this information and communication age, particularly in the areas of data mining (DM) and database management systems (DBMS). With the huge data sets, now the traditional market forecasting techniques are becoming of no use. Several statistical techniques, like multivariate analysis, time series and so on, are also failing to perform satisfactory clustering or segmentation. In this connection, a new form of knowledge management technologies with soft computing and hard computing like data mining, machine learning, artificial intelligence, etc. will definitely solve market-related problems [5].

In this competitive world, today, most sellers want to know the needs and preferences of the customer. Now they profusely maintain good relationships with customers at every stage of business operations. The concept of maintaining a good relationship with the customer is known as customer relationship management (CRM). This theory of customer relationship management is becoming an integral part of marketing strategy. With the proliferation of the Internet, the idea of relationship management has become popular due to several computational approaches. The company and customers can easily interact and understand each other by learning the hidden knowledge from the enormous quantity of data. The concept of understanding and analysing the hidden knowledge of the customer is data mining. Data

mining is a computational analysis process that discovers the consumer's taste and preferences through customer segmentation, dividing huge sets of data [6]. The data mining approach is also useful for manufacturers who have lost their quality when the products decay. In this case, the recency, frequency, and monetary (RFM) form of segmentation failed to quantify the exact preference rather than other methods like the Fuzzy Analytic Network Process (FANP) [7]. Sometimes, data mining techniques are useful for profiling the customer base, targeting, aligning the right channels, cross-selling products, enhancing customer relationships and providing value to the customer [8]. However, prioritising the customer within the existing customer base is also an important technique in data mining. To improve the service quality and effectiveness of the product, importance-performance analysis (IPA) is also a part of data mining [9]. Customer segments are highly volatile; they may change according to the preference of the customer, which creates confusion about the re-computation of data. These uncertainties require streaming of data in a proper form where data mining helps to cluster the data. As a result, customer segmentation performs continuously [10]. Data mining techniques are also predicting the future probability and behaviours that allow businesses to be more practical and knowledge-driven [11]. Data mining techniques also provide the advantage of customer segmentation functions [12]. Data mining also classifies blogs into supervised and unsupervised learning models for extracting knowledge from voice over the Internet protocol [13].

After a meticulous review of 550 academic literature, 57 research articles and 17 conference papers were considered in this review process. This paper discusses customer segmentation via data mining techniques from a review perspective. This paper is a systematic investigation into supervised, unsupervised and other data mining techniques. The supervised approaches, such as neural networks, naive Bayes, linear regression, logistic regression, support vector machine (SVM), K-nearest neighbour, boosting and decision tree (DT), hidden Markov model (HMM), and random forest have an enormous contribution to object detection and classification. In unsupervised approaches, complex classification of data, identification and processing of variables have more emphasis through K-means clustering, K-nearest neighbours (KNN), hierarchal clustering, anomaly detection, neural networks, principle component analysis, independent component analysis, apriori algorithm, etc. Some of the research articles on other data mining techniques, such as chi-square automatic interaction detector (CHAID), RFM, genetic algorithm (GA), and logistic regression, etc., have revealed classification and relationship management. The paper is organised into 5 sections. Section 2 presents various issues involved with customer segmentation. Section 3 explains various segmentation techniques. Section 4 is about critical investigation. Section 5 concludes the discussion and conclusion.

2 Various Issues Involved with Customer Segmentation

Consumers have different needs and expectations as per their characteristics. In consumer behaviour research literature, we can observe several segmentation variables, such as demographic, geographic, psychographics, decision-making, behavioural, purchase behaviour, personality, lifestyle, situation factors, etc. However, from a broader perspective, the researcher classified the customer segmentation into four major areas, such as geographic characteristics, demographic profile, psychographic profile, and behavioural aspects. On the other hand, some researchers have classified it into two distinct forms. They are observed and unobserved variables. The observed variables, in general, are geographic, demographic, and socio-economic, whereas purchase frequency and customer loyalty are considered as product-specific or brand-specific variables. Sometimes, variables like lifestyle and psychographics are unobserved in general and product benefits, intention, preference, etc. are considered product-specific [5]. So, customer segmentation is an emerging area of research that has several issues replicating consumer behaviour on a product or brand. In the decision-making process, customer segmentation is an integral part of the marketing strategy which builds customer relationships, segregates customers into different groups, and provides different facilities in the niche market. In particular, for mobile users, it develops VIP customer segmentation which can easily identify their needs and facilitate the service [14].

The rapid development of computer technologies across the globe has changed the tastes of telecom subscribers. Now it is high time for a telecom company to understand the characteristics of the consumer to provide distinct services. Segmentation is the only way to cluster customers into different bases and provide the service to attract and retain customers [15]. Customer segmentation is also important in the retail sector today. With the huge quantity of customer data, a retail firm may not be able to keep the customer informed. So, data mining techniques could help to mine the data among lost customers and help the retailer to build customer relationships [16]. In this regard, customer segmentation will provide a wealth of information about customers. Customer segmentation is the strategic resource for an enterprise to gain competitive advantages and make customers profitable [17]. Segmentation is important for providing customer lifetime value (LTV). But now the statement has become vague due to a lot of competition. Therefore, customer values like current value, potential value, and customer loyalty will be an important asset for any marketer to understand the customer better [18]. Customer segmentation is classifying the value via the RFM model and rough set theory (RST) theory to understand the customer and maintain the relationship [19]. According to the previous literature, segmentation has various critical issues like problem recognition, design of the research, data collection, data analysis, and implementation [5]. Table 1 depicted major issues related to customer segmentation issues to counter the problems.

Customer segmentation offers a tactic decision for supporting services and profitability for businesses. It supports all kinds of business decisions for financial growth and development. Therefore, making a good customer segmentation method is a

Table 1 Issues related to customer segmentation

Issues of customer segmentation	Major consideration
Recognize the problem	• Segmentation concept • Information related to customer • Classification of the variables • Customer segmentation base selection • Finance and other limitations
Design the research	• Collection of data • Instrument validity • Objectives of segmentation • Stability of variables
Data collection	• Source of data
Data analysis	• Data analysis and classification of segmentation • Clustering data sets • Reliability and validity of information
Implementation	• Implement on target customer • Select segments

systematic way of defining the tools that help the business to grow and develop. Therefore, selection of the right tools involves cross-functional cause to deal with the business goal. Customer segmentation has a lot of pros and cons while classifying customers into different profiles. Sometimes it procures, retains, and attracts the customer. It clusters the customers according to the market demand. However, it could be successful when accurate data interpretation, knowledge discovery, and information dissemination are properly done. Often, due to inexact information, it is not effective. The manual process of segmentation is time-consuming, un-scalable, and not agile. Therefore, segmentation does not help one-to-one marketing. With the help of the latest technologies like data mining, artificial intelligence, machine learning, etc., accurate segmentation is possible and makes the customer profitable.

3 Segmentation Techniques

In general, customer segmentation involves a broad variety of techniques like cluster analysis [10], cluster-wise-regression, AID/CHAID, multiple regression, discrimination analysis, latent class structure, inductive learning techniques, soft computing techniques [5], and data mining (the detailed theory proposed in the next section) are used in different market conditions. However, it is difficult to classify the group of customers according to their attributes. We have to consider the classical method. In the classical theory, some researchers gave importance to the data preparation framework and data analysis framework, which include supervised, unsupervised

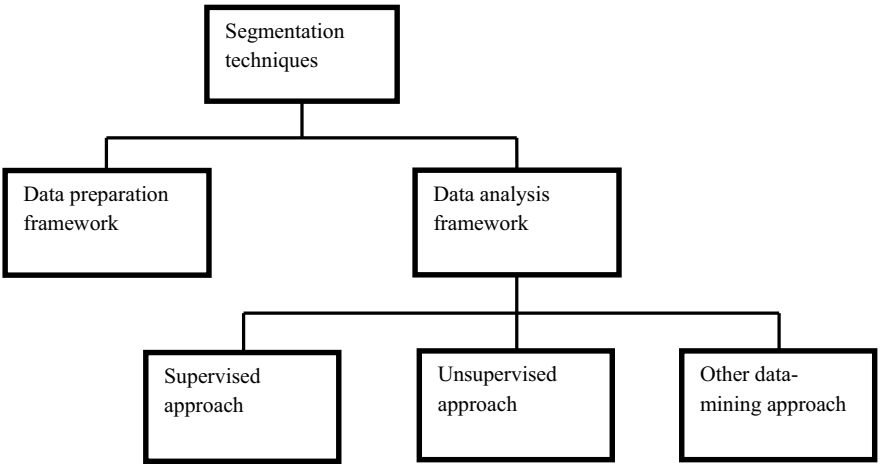


Fig. 1 Customer segmentation techniques

and other methods of data mining approach (Fig. 1). Most of the techniques related to artificial neural networks (ANNs), fuzzy logic (FL), machine learning (ML), RST and evolutionary methods (EM) such as GA are the main data mining tools to analyse data perfectly. These technologies have been widely used in data preparation and data analysis. It is a challenging task for modern marketing professionals to consider the right technique or algorithm. Most of these algorithms have significant advantages and disadvantages also. To avoid this problem, researchers should consider either a supervised or unsupervised approach. The supervised approach is a classification method where the inputs and outputs are mapped properly. In the supervised approach, all the common algorithms, i.e. support vector machines, logistic regression, artificial neural networks, naive Bayes, and random forests, significantly work further. These approaches follow a hierarchical process to maintain a good relationship between input and output datasets. The unsupervised approaches are clustering of data inherently. Some familiar algorithms include k-means clustering, principal component analysis, and auto encoders. Since no labels are provided, there is no specific way to compare model performance in most unsupervised approaches. In this connection, DM techniques using neural networks, decision trees, genetic algorithms, fuzzy logic, and K-nearest neighbour could be able to predict, comprehend, and cluster the customers properly [20]. Besides the non-traditional methods, some traditional techniques like self-organizing maps (SOM) can also be used to make segmentation. In this approach, a set of initial cluster prototypes are made before applying the K-means to get the final clusters of data sets through near visualization. Some researchers said that the U-matrix is also one of the best options for clustering the data for analysing the results by time of hits.

3.1 Data Preparation Framework

Data preparation is a systematic way of transforming raw data into a basic form of data for predictive analysis to remove errors or mistakes. Data preparation is a challenging task to acquire proper prediction analysis. It uses automatic search like grid and random search to find unity in data preparation. Often it is difficult to gather a variety of data. For example, the data might be stored in a CSV file for classification and regression consists of rows, columns, and values for any data preparation method. However, most of the authors articulated that data preparation techniques are inferred using statistical and non-statistical techniques. Statistical techniques like exploratory factor analysis and correspondence analysis; and computational techniques such as soft computing tools (e.g. RST or GA) are typically used in data preparation [5]. Exploratory factor analysis (EFA) is a common statistical method applicable to multivariate statistics to uncover a relatively large set of data. Most of the time, researchers use this technique for scaling the data sets through the questionnaire. EFA is accurate as each factor is symbolized by multiple measured variables. EFA is based on common factors, unique factors, and errors of measurement. With this EFA model, we can easily identify the common factors and other related manifest variables. The correspondence analysis (CA) is an expansion of principal component analysis appropriate for discovering relationships amongst qualitative variables (or categorical data). Like principal component analysis, it also offers a solution for summarizing and visualizing the data in two-dimension plots. Correspondence analysis is a significant form of geometric approach for visualizing rows and columns of a two-way contingency table appropriately. The main aim of this tabular form is to provide a global view of the data for easy interpretation. However, these statistical techniques have been replaced by soft computing to segment or classify the data and provide accurate results. In particular, soft computing (SC) is an improved technique over conventional traditional systems. It is also part of hard computing. It has many intelligent and user-friendly features. Soft computing consists of FL, ANNs, RST, and EM. The principal component of soft computing is to eliminate the uncertainty and vagueness of data through fuzzy tools and EM, which are involved in the optimization and searching process. Furthermore, ANNs and RST will solve the classification and rule generation problems. Recently, soft computing technologies have been used for resolving data mining problems. Soft computing is widely used for the analysis and interpretation of data. RST is mathematical computation and granular approximation which discovers the hidden pattern in an uncertain environment widely used in soft computing. Therefore, soft computing is a computational method that is useful for data preparation.

3.2 Data Analysis Framework

Segmenting the customer into different groups, such as geography, demography, psychographic, and behavioural, is an easy form of classification of customer data to analyse the customer's needs and expectations. There are various approaches to classifying the market into different groups, popularly known as cluster analysis. In an article, Calantone and Johar [21] proposed that cluster analysis could classify customer data explicitly. They proposed that benefits customers should be analysed properly in the tourism industry, where the marketing strategy formulation such as understanding customers, product positioning, advertising copy testing, and new market development will help to establish the market. However, the analysis used by statistical analysis like factor analysis may extend the resultant output. In this context, computational approaches like supervised, unsupervised and other data mining approaches are widely used for data analysis.

3.2.1 Supervised Approach

A supervised approach is a systematic application of artificial intelligence (AI) where a computer algorithm is absolutely trained on input data for assumed output. It creates labelled data according to the specific question asked by the customer. The supervised approach is also the finest learning approach for machine learning, useful for forecasting financial results, identifying fraud, recognizing objects in images, and also evaluating risk. In a supervised approach, the input and output data are known in advance for better prediction with the appropriate classification. Object detection is one of the important aspects of the supervised approach to computer vision. The classical object detection approaches, such as background subtraction and saliency detection, do not have manual collection and labelling of samples. Generally, they do not train the samples for the classification of labelled data like the supervised approach. But sometimes it is absolutely affected by noise issues like changes in luminance and the cluttered background. On the other hand, supervised approaches like support vector machine, boosting and decision tree have good performance in object detection. But it needs a substantial human interface to train the data for labelling. In this connection, Wang et al. [22] developed a model to avoid manual detection of objects or videos where the extension of the boosting algorithm (soft label boosting) will help to train the samples with a soft (probabilistic) label in place of a hard (binary) label. Tracking the emotions in the images or video clips is also an important feature of the supervised approach.

In their paper, Malandrakis et al. [23] proposed an emotion tracking system in the movie where the valance-arousal scale was detected through a continuous annotated database. A supervised approach is proposed in their paper using hidden Markov models in each dimension. They used HMMs to predict arousal and valance features in the movie. They found that the sensor could be captured microscopically and detect emotions. However, evaluation of the supervised approach is also important

for image segmentation with the use of a proper algorithm [24]. Sentiment analysis (SA) is a newly emerged research topic which unlocks a new future for businessmen, writers, and bloggers. It is an emerging form of computational algorithm to understand the percentage of product acceptance and rejection where the business acumen builds up their strategy to improve the product performance. In this regard, opinion mining will be possible to find the exact intention of the customer through supervised machine learning models [25]. The supervised approach also detects the musical boundaries between verse and chorus segments. Here the perceptual aspects such as timbre, harmony, melody, and the rhythm of music through boosting [26]. Graph base spectral algorithm is a recent topic in research today, which detects image objects through a clustering algorithm in a meaningful enlarge structure [27]. The fault diagnosis system (FDS) is also an improved method of supervised learning using a support vector machine for appropriate decision-making [28]. The decomposition of nuclear waste objects through robotics is a matter of concern where the RGBD-based detection and categorization is applied by a deep convolutional neural network (DCNN) from unlabelled RGBD videos. It helps to make an object detection benchmark to recognize waste objects perfectly [29]. In this connection, supervised learning is a leading algorithm that was developed to identify the data, cluster and recognize to perceive the individual customer expectations. This type of segmentation will be helpful for researchers and business leaders to develop the product quality and meet the needs of the customers.

3.2.2 Unsupervised Approach

An unsupervised approach is a form of an algorithm that learns patterns from unlabelled data. In particular, it captures patterns such as neural prediction or prospect density. It develops imaginative content through the internal representation of data. Unlike the supervised approach, it has no human interaction, rather segmentation of data by neural network and probabilistic method. It finds an interesting pattern from various unlabelled sensor data without prior information. One of the popular techniques of the unsupervised approach is data mining for the activity recognition task. Though it has no human interaction, the classification of complex data is possibly effective in the customer segmentation process through pattern recognition. Often, data sets have larger features and fewer occurrences are a relatively challenging task for machine learning. However, with these multiple features of data sets, there may be irrelevant or redundant information that causes damage in terms of correctness or training time. To deal with these complex situations, the feature selection (FS) and feature discretization (FD) methods will be helpful to recognize the data sets. In particular, in the pre-processing stage, some classification algorithms deal with discrete features where the FD technique finds the representation of each feature. On the other hand, FS is aiming at dropping features to target the curse of dimensionality problems, often permitting learning algorithms to be better-performing classifiers. Therefore, feature discretization-based algorithms could reduce the redundancy and classify the data set [30]. In an unsupervised approach, fuzzy-based clustering is

evaluated through the fuzzy joint points (FJP) method where the data set is classified in hierarchical order [30].

DNA array analysis is a functional algorithm to measure the expression of multiple genes in an unsupervised approach. Just like supervised learning, a two-way clustering framework is also able to identify gene patterns and perform cluster discovery on samples where connectivity among the groups of genes could be possible [31]. Speech recognition and grouping of voices through co-channel (two-talker) speech separation is also a part of the unsupervised learning approach. For voice segregation and segmentation of speech, a differential algorithm like tandem will work to separate the unvoiced speech [32]. This unsupervised approach is also applied for the summarization of opinions. The state-of-the-art algorithm has been used in this process where the summarization method is informative and readable [33]. This approach also detects human activity recognition from raw data by wearable sensors to identify expectations [34]. The segmentation of data classification could be possible through multidimensional time series using the hidden Markov model, which predicts human activity accurately. Automatic summarization of documents is a recent development in the summarization of documents where the algorithms classify the data into words, sentences, and phrases and finally process the document. It also observes the relevancy, redundancy, and length of the document while summarizing it [35]. Most researchers used the unsupervised learning approach for different perspectives, such as facial landmark detectors, protocol features of word extraction, product attribute extraction, clusters of pixel images, and so on.

3.2.3 Other Data Mining Approaches

In recent years, customer segmentation in direct marketing has become more effective with the development of database marketing techniques. These types of data mining approaches ensure direct marketers segment customers in a better way to perform with a different marketing strategy. The data mining approaches such as CHAID, RFM, GA, and logistic regression were used as the analytical tools for direct marketing segmentation with two types of data sets. It was found that amongst all the approaches, RFM is the perfect approach. However, CHAID is also an optimal solution for segmenting the data into sequence. So an empirically based RFM approach could replace both CHAID and logistic regression in database marketing systems [36]. Therefore, it can be observed from several studies that RFM technology has been used vividly to segment customers to access information. The marketing representatives of commercial banks can segment through k-means classification to obtain potential customers. To obtain useful information from the customer, four types of data mining methods, such as neural network, C5.0, classification and regression tree, and chi-squared automatic interaction detector, will definitely be helpful to detect the background information for credit card holders [37, 38]. Market segmentation has a key role in continuing the relationship with a loyal customer. In this regard, there must be a correlation between the retailer and the customer. By the use of the

divisive cluster analysis technique of data mining, the retailer can find all kinds of information from the customer database [39].

The advent of technology for data optimization and screening is an important technique for data mining that mines vast data sets and classifies the market accordingly. In particular, ANN and particle swarm optimization (PSO) methods are recent developments for market decision strategy. With the integration between statistical analysis and particle swarm optimization, we can reduce redundant data and segment the market properly [40]. Data mining techniques have become an indispensable method in market segmentation. The classification of larger data sets from databases is a recent form of market research where some intelligent solutions, such as neural networks, evolutionary algorithms (EA), fuzzy theory, RFM, hierarchical clustering, K-means, bagged clustering, kernel methods, Taguchi method, multidimensional scaling, model-based clustering, rough sets, and others, will be very effective and time-bound [41]. So, clustering the data is an important feature of data mining techniques where latent class analysis (LCA), prior clustering, and some description of similarity or distance measures of data are used for segmenting large groups of customers for individual expectations [42]. To understand the various research articles, we can confirm that data mining is vividly used for the exploration and prediction of expected outcomes in the heterogeneous market. Data mining is used for classification, clustering, association, and sequential analysis. In this regard, certain statistical applications such as regression, time series, association and sequential analysis will be beneficial for mining large data sets [43].

4 Critical Investigation

Customer segmentation is an integral approach to target the customer and position the brand in the mindset of the customer. Though there are several approaches, such as supervised, unsupervised, and other data mining techniques that have been used since 1990 by various researchers at different points of time, it has become a part of customer segmentation to classify and cluster large data. In this paper, we have extracted articles from various online bibliographies of academic articles on customer segmentation, such as ABI/INFORM database, Science Direct, Emerald, IEEE Transactions, JOSTER, Springer, Google scholar, and Wily online library. The academic articles are searched for keywords like customer segmentation, market segmentation, and customer segmentation and data mining. Among 550 articles, the relevant literature on customer segmentation using data mining techniques has been considered as a state-of-the-art review. In this paper, we considered almost 57 articles and 17 conference papers. After detailed observation of the literature, it found that data mining techniques like K-means, RFM, GA, and other algorithms are used in research for classification of large data sets to target customers and create meaningful marketing strategy.

4.1 Impact of Segmentation Variables

Consumers have an extensive variety of characteristics. Based on their variables, we can find several segmentation variables, such as geographic, demographic, firm graphics, decision-making processes, situational factors, personality, profitability, benefits sought, and so on. According to Kotler and Keller [44], segmentation variables are classified into four important areas, such as demographic features, geographic characteristics, psychographic and behavioural variables. On the other hand, several authors have articulated the levels of variables, e.g. general variables, domain-based and brand-specific; and the objectivity-oriented and subjectivity of the variables. The number of variables that have been developed over different periods has faced a massive challenge. Too many have been proposed to make it practical for the market to empirically compare them all when trying to segment a market. In this regard, the classification can be broadly divided into general observed variables (e.g. geographic features, demographic profile, socio-economic variables) and unobserved variables (e.g. lifestyle and psychographics); product-oriented observed variables (e.g. usage frequency and loyalty) and product-oriented unobserved variables (e.g. benefits, preferences, and intentions) [45]. Therefore, the selection of the proper segmentation variable is a significant point to consider. In his article, wind [46] articulated that most of the segmentation studies were on consumer goods. However, the process of segmentation is also applicable in the industrial market. So, before selecting an appropriate segmentation method, we must think about the problems and prospects of segmentation. To select the proper method, the priori segmentation design and cluster-based design are most essential. In prior segmentation designs, the marketer was able to segment through product purchase, loyalty, and type of customer wherein cluster-based design segments determine the benefits, needs, and attitudes of customers. Further, the advantages and disadvantages of segmentation are also necessary. After observing several academic literatures, we found that there is an equally importance on variety of segmentation models. But we must be careful to select the segmentation method based on management's specific objectives and also on current trends in the consumer market (Table 2; Fig. 2).

4.2 Model Reliability in Segmentation

Despite the importance of segmentation analysis on different data sets, minor attention has to be paid to check the reliability and validity. Because some variables, like demographics (age, gender, income, religion, etc.) are more reliable than behavioural or psychological characteristics. In particular, in the case of an attitude survey, proper care should be considered and should test the reliability of data. To check the reliability of data, statistical measures like factor analysis, conjoint analysis, co-relation, component matrix, etc. will be beneficial for data analysis. However, these traditional methods could not provide accuracy due to several exceptions to the number

Table 2 Impact study of segmentation variables

Types of segmentation	Focus area	References
General observable variables	Demographics	[47–50]
	Socio-economic	[51, 52]
	Behavioural	[53, 54]
	Cultural	[55, 56]
General unobservable variables	Lifestyle	[50, 57–60]
	Psychographic	[61, 61–64]
Product specific variables	Usage frequency	[65]
	Loyalty	[66–68]
Product specific unobservable variables	Benefits	[69, 70]
	Attitude	[71, 72]

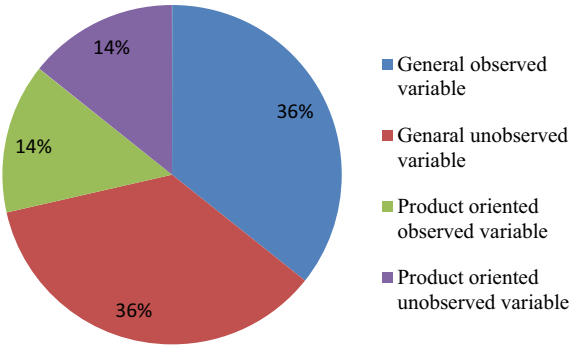


Fig. 2 Types of customer segmentation variables

of items. In this connection, perceptual studies like a generalization of data could provide better analytical results. Therefore, there is a need for instrument development in data reliability [46]. Commonly, there are two potential approaches to measure the reliability, such as degree of consistency and cross-validation [73]. The former approach can be executed through clustering or classification of data sets, which requires multiple time verification. The latter approach can be performed by dividing the data into two different parts and performing the analysis to check the reliability of the sample parts. When the clustering process is executed, the latter method can be modified by obtaining the cluster centroids from the first part and using them to describe clusters in the second part. Cross validation is a more generalized approach compared to the first approach. Concerning cross-validation of the data discriminate measure of the Wilk's Lambda (k) and the Kappa, the index is the most famous method applied in marketing research [74]. Before examining the experiential task, we will immediately believe whether any type of reliability has been taken into account or not. The distance between the clusters should be measured through squares within and between the clusters, a scatter matrix of data points, and indexes. Further, different indexes could be employed to determine the number of

fuzzy clusters in the datasets. Some of the indexes also compare the clusters. Hence, inherently, the data sets should be checked and rechecked through the proper method to test their reliability.

4.3 Selection of Proper Data Mining Model

Data mining is the significant procedure of analysing large volumes of data to ascertain business acumen, which helps companies to resolve problems, mitigate risks, and grasp new opportunities. This particular division of data science derived from the similarities in data between searching for important information in a large database and mining a peak. Both processes need sifting through wonderful amounts of material to find hidden value. Data mining can answer all kinds of business questions that traditionally took more time to resolve manually. Using a wide range of statistical techniques to analyse data from a different perspective, users can identify patterns, trends, and relationships. Customer segmentation is a measure of concern for market analysis where proper data classification is important. Though there is the application of several statistical techniques in a customer database, data mining techniques could help predict, analyse and profile the customer in a significant way. Several academic literature has given the importance of various data mining techniques, like supervised, unsupervised, and other data mining techniques, but it could be difficult to identify the exact data mining techniques for their study. So the researchers should have domain knowledge of business, techniques, and also a fitness model. Here, we proposed a data mining model (Fig. 3) based on the suitability of customer needs and expectations.

5 Discussion and Conclusion

Customer segmentation using data mining is a recent study where most of the academic literature suggests the classification of data. Some of these studies emphasized different clustering methods also. However, the selection of segmentation techniques is a challenging task for a business concern. About the selection of the segmentation, we must think about two important aspects, i.e. the objective of management and the recent trends in the market. The classical methods like factor analysis, regression, conjoint analysis, or co-efficient determinants may not provide accurate predictions. Therefore, in this review, we observed that computational algorithms could justify businessmen for analysis and prediction. As we know, most business acumen are expanding their products or services into different markets and also searching for a better customer portfolio where they can target customers and position their brands. In this connection, we highlighted four types of segmentation techniques, such as general observable variables, unobservable variables, product-specific observable variables, and product-specific unobservable variables. In the first case, the variables

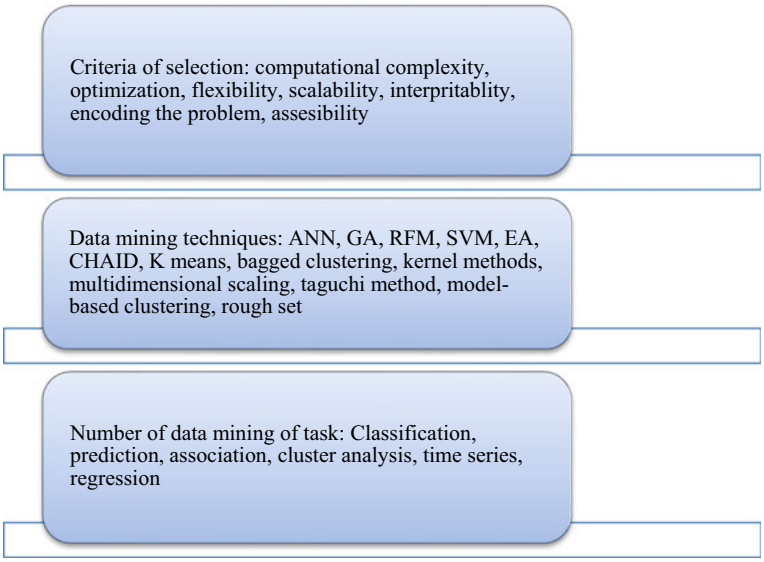


Fig. 3 Proposed model for selection of data mining techniques

are geographic, demographics, socio-economic, and culture; in the second case, they include lifestyle, psychographics, attitude, and emotions; in the third case, the variables are frequency of purchase and loyalty; and finally, in the fourth case, the variables are benefits, preference, and intention. Therefore, segmenting the customer through data mining techniques like K-mean, RFM, GA, ANN, kernel method, PSO could be helpful to the marker to segment properly.

In the future, the marketing strategy will rely on these customer segmentation techniques with a large data bank. For example, a credit card service provider will collect all kinds of customer information from the bank and facilitate the credit card. An insurance company is collecting prospect customer information to sell its services. Though the data sets are large, little human interaction is also necessary for prediction. Data mining techniques use algorithms to quantify the labelled and unlabelled training inputs for a valid output. So, the supervised and unsupervised approaches will justify the adequate output. With the use of data mining techniques, the business will grow with a stringent marketing strategy to expand and diversify the product or services.

References

1. G. Lefait, T. Kechadi, Customer segmentation architecture based on clustering techniques, in *2010 Fourth International Conference on Digital Society* (IEEE, 2010). https://doi.org/10.1109/ICDS.2010.47

2. P.Q. Brito et al., Customer segmentation in a large database of an online customized fashion business. Robot. Comput.-Integr. Manuf. **36**, 93–100 (2015). https://doi.org/10.1016/j.rcim.2014.12.014
3. W.R. Smith, Product differentiation and market segmentation as alternative marketing strategies. J. Mark. **21**(1), 3–8 (1956). https://doi.org/10.1177/002224295602100102
4. A. Nairn, P. Berthon, Creating the customer: the influence of advertising on consumer market segments—evidence and ethics. J. Bus. Ethics **42**(1), 83–100 (2003). https://doi.org/10.1023/A:1021620825950
5. A. Hiziroglu, Soft computing applications in customer segmentation: state-of-art review and critique. Expert Syst. Appl. **40**(16), 6491–6507 (2013). https://doi.org/10.1016/j.eswa.2013.05.052
6. A. Hajiha, R. Radfar, S.S. Malayeri, Data mining application for customer segmentation based on loyalty: an Iranian food industry case study, in *2011 IEEE International Conference on Industrial Engineering and Engineering Management* (IEEE, 2011). https://doi.org/10.1109/IEEM.2011.6117968
7. V. Golmah, G. Mirhashemi, Implementing a data mining solution to customer segmentation for decayable products—a case study for a textile firm. Int. J. Database Theory Appl. **5**(3), 73–90 (2012)
8. M.M.T.M. Hassan, M. Tabasum, Customer profiling and segmentation in retail banks using data mining techniques. Int. J. Adv. Res. Comput. Sci. **9**(4), 24–29 (2018)
9. S.Y. Hosseini, A.Z. Bideh, A data mining approach for segmentation-based importance-performance analysis (SOM–BPNN–IPA): a new framework for developing customer retention strategies. Serv. Bus. **8**(2), 295–312 (2014). https://doi.org/10.1007/s11628-013-0197-7
10. M. Carnein, H. Trautmann, Customer segmentation based on transactional data using stream clustering, in *Pacific-Asia Conference on Knowledge Discovery and Data Mining* (Springer, Cham, 2019). https://doi.org/10.1007/978-3-030-16148-4_22
11. W. Wang, S. Fan, Application of data mining technique in customer segmentation of shipping enterprises, in *2010 2nd International Workshop on Database Technology and Applications* (IEEE, 2010). https://doi.org/10.1109/DBTA.2010.5659081
12. J. Ranjan, R. Agarwal, Application of segmentation in customer relationship management: a data mining perspective. Int. J. Electron. Custom. Relat. Manag. **3**(4), 402–414 (2009). https://doi.org/10.1504/IJECRM.2009.029298
13. L.-S. Chen, C.-C. Hsu, M.-C. Chen, Customer segmentation and classification from blogs by using data mining: an example of VOIP phone. Cybern. Syst. Int. J. **40**(7), 608–632 (2009). https://doi.org/10.1080/01969720903152593
14. Z. Yihua, Vip customer segmentation based on data mining in mobile-communications industry, in *2010 5th International Conference on Computer Science & Education* (IEEE, 2010). https://doi.org/10.1109/ICCSE.2010.5593669
15. C. Qiuru et al., Telecom customer segmentation based on cluster analysis, in *2012 International Conference on Computer Science and Information Processing (CSIP)* (IEEE, 2012). https://doi.org/10.1109/CSIP.2012.6309069
16. H. Gong, Q. Xia, Study on application of customer segmentation based on data mining technology, in *2009 ETP International Conference on Future Computer and Communication* (IEEE, 2009). https://doi.org/10.1109/FCC.2009.66
17. X. Lai, Segmentation study on enterprise customers based on data mining technology, in *2009 First International Workshop on Database Technology and Applications* (IEEE, 2009). https://doi.org/10.1109/DBTA.2009.96
18. H. Hwang, T. Jung, E. Suh, An LTV model and customer segmentation based on customer value: a case study on the wireless telecommunication industry. Expert Syst. Appl. **26**(2), 181–188 (2004). https://doi.org/10.1016/S0957-4174(03)00133-7
19. C.-H. Cheng, Y.-S. Chen, Classifying the segmentation of customer value via RFM model and RS theory. Expert Syst. Appl. **36**(3), 4176–4184 (2009). https://doi.org/10.1016/j.eswa.2008.04.003

20. S. Kelly, Mining data to discover customer segments. Interact. Mark. **4**(3), 235–242 (2003). https://doi.org/10.1057/palgrave.im.4340185
21. R.J. Calantone, J.S. Johar, Seasonal segmentation of the tourism market using a benefit segmentation framework. J. Travel Res. **23**(2), 14–24 (1984). https://doi.org/10.1177/004728758402300203
22. W. Wang et al., A weakly supervised approach for object detection based on soft-label boosting, in *2013 IEEE Workshop on Applications of Computer Vision (WACV)* (IEEE, 2013). https://doi.org/10.1109/WACV.2013.6475037
23. N. Malandrakis et al., A supervised approach to movie emotion tracking, in *2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (IEEE, 2011). https://doi.org/10.1109/ICASSP.2011.5946961
24. L. Yang et al., A supervised approach to the evaluation of image segmentation methods, in *International Conference on Computer Analysis of Images and Patterns* (Springer, Berlin, Heidelberg, 1995). https://doi.org/10.1007/3-540-60268-2_377
25. Md.S. Islam et al., Supervised approach of sentimentality extraction from Bengali Facebook status, in *2016 19th International Conference on Computer and Information Technology (ICCIT)* (IEEE, 2016). https://doi.org/10.1109/ICCITECHN.2016.7860228
26. D. Turnbull et al., A supervised approach for detecting boundaries in music using difference features and boosting, in *ISMIR* (2007)
27. L. Yang et al., A supervised approach to the evaluation of image segmentation methods, in *International Conference on Computer Analysis of Images and Patterns* (Springer, Berlin, Heidelberg, 1995). https://doi.org/10.1016/j.neucom.2011.09.002
28. I. Monroy et al., A semi-supervised approach to fault diagnosis for chemical processes. Comput. Chem. Eng. **34**(5), 631–642 (2010). https://doi.org/10.1016/j.compchemeng.2009.12.008
29. L. Sun et al., A novel weakly-supervised approach for RGB-D-based nuclear waste object detection. IEEE Sens. J. **19**(9), 3487–3500 (2018). https://doi.org/10.1109/JSEN.2018.2888815
30. A.J. Ferreira, M.A.T. Figueiredo, An unsupervised approach to feature discretization and selection. Pattern Recogn. **45**(9), 3048–3060 (2012). https://doi.org/10.1016/j.patcog.2011.12.008
31. E.N. Nasibov, G. Ulutagay, A new unsupervised approach for fuzzy clustering. Fuzzy Sets Syst. **158**(19), 2118–2133 (2007). https://doi.org/10.1016/j.fss.2007.02.019
32. Ke. Hu, D.L. Wang, An unsupervised approach to cochannel speech separation. IEEE Trans. Audio Speech Lang. Process. **21**(1), 122–131 (2012). https://doi.org/10.1109/TASL.2012.2215591
33. K. Ganesan, C.X. Zhai, E. Viegas, Micropinion generation: an unsupervised approach to generating ultra-concise summaries of opinions, in *Proceedings of the 21st International Conference on World Wide Web* (2012)
34. D. Trabelsi et al., An unsupervised approach for automatic activity recognition based on hidden Markov model regression. IEEE Trans. Autom. Sci. Eng. **10**(3), 829–835 (2013). https://doi.org/10.1109/TASE.2013.2256349
35. R.M. Alguliyev, R.M. Aliguliyev, N.R. Isazade, An unsupervised approach to generating generic summaries of documents. Appl. Soft Comput. **34**, 236–250 (2015). https://doi.org/10.1016/j.asoc.2015.04.050
36. J.A. McCarty, M. Hastak, Segmentation approaches in data-mining: a comparison of RFM, CHAID, and logistic regression. J. Bus. Res. **60**(6), 656–662 (2007). https://doi.org/10.1016/j.jbusres.2006.06.015
37. W. Li et al., Credit card customer segmentation and target marketing based on data mining, in *2010 International Conference on Computational Intelligence and Security* (IEEE, 2010). https://doi.org/10.1109/CIS.2010.23
38. Z. Lu et al., Customer segmentation algorithm based on data mining for electric vehicles, in *2019 IEEE 4th International Conference on Cloud Computing and Big Data Analysis (ICCCBDA)* (IEEE, 2019). https://doi.org/10.1109/ICCCBDA.2019.8725737

39. V.L. Miguéis, A.S. Camanho, J. Falcão e Cunha, Customer data mining for lifestyle segmentation. Expert Syst. Appl. **39**(10), 9359–9366 (2012). https://doi.org/10.1016/j.eswa.2012.02.133
40. C.-Y Chiu et al., An intelligent market segmentation system using k-means and particle swarm optimization. Expert Syst. Appl. **36**(3), 4558–4565 (2009). https://doi.org/10.1016/j.eswa.2008.05.029
41. S. Dutta, S. Bhattacharya, K.K. Guin, Data mining in market segmentation: a literature review and suggestions, in *Proceedings of Fourth International Conference on Soft Computing for Problem Solving* (Springer, New Delhi, 2015). https://doi.org/10.1007/978-81-322-2217-0_8
42. E.R. Swenson, N.D. Bastian, H.B. Nembhard, Healthcare market segmentation and data mining: a systematic review. Health Mark. Q. **35**(3), 186–208 (2018). https://doi.org/10.1080/07359683.2018.1514734
43. S. Mckechnie, Integrating intelligent systems into marketing to support market segmentation decisions. Intell. Syst. Account. Finance Manag. Int. J. **14**(3), 117–127 (2006). https://doi.org/10.1002/isaf.280
44. P. Kotler, K.L. Keller, *Marketing Management*, ed. by W. Lassar, international 11th edn. (Prentice Hall, New Jersey, 2003)
45. M. Wedel, W.A. Kamakura, *Market Segmentation: Conceptual and Methodological Foundations*, vol. 8 (Springer Science & Business Media, 2012)
46. Y. Wind, Issues and advances in segmentation research. J. Mark. Res. **15**(3), 317–337 (1978). https://doi.org/10.1177/002224377801500302
47. L. Alfansi, A. Sargeant, Market segmentation in the Indonesian banking sector: the relationship between demographics and desired customer benefits. Int. J. Bank Mark. (2000). https://doi.org/10.1108/02652320010322976
48. D.G. Tonks, Validity and the design of market segments. J. Mark. Manag. **25**(3–4), 341–356 (2009). https://doi.org/10.1362/026725709X429782
49. M. Taks, J. Scheerder, Youth sports participation styles and market segmentation profiles: evidence and applications. Eur. Sport Manag. Q. **6**(2), 85–121 (2006). https://doi.org/10.1080/16184740600954080
50. J. Bruwer, E. Li, Wine-related lifestyle (WRL) market segmentation: demographic and behavioural factors. J. Wine Res. **18**(1), 19–34 (2007). https://doi.org/10.1080/09571260701526865
51. P. Vyncke, Lifestyle segmentation: from attitudes, interests and opinions, to values, aesthetic styles, life visions and media preferences. Eur. J. Commun. **17**(4), 445–463 (2002). https://doi.org/10.1177/0267323102017004030 1
52. A. Vellido, P.J.G. Lisboa, K. Meehan, Segmentation of the on-line shopping market using neural networks. Expert Syst. Appl. **17**(4), 303–314 (1999). https://doi.org/10.1016/S0957-4174(99)00042-1
53. J. Swait, A structural equation model of latent segmentation and product choice for cross-sectional revealed preference choice data. J. Retail. Consum. Serv. **1**(2), 77–89 (1994). https://doi.org/10.1016/0969-6989(94)90002-7
54. T. Teichert, E. Shehu, I. von Wartburg, Customer segmentation revisited: the case of the airline industry. Transp. Res. Part A Policy Pract. **42**(1), 227–242 (2008). https://doi.org/10.1016/j.tra.2007.08.003
55. A. Lindridge, S. Dibb, Is 'culture' a justifiable variable for market segmentation? A cross-cultural example. J. Consum. Behav. Int. Res. Rev. **2**(3), 269–286 (2003). https://doi.org/10.1002/cb.106
56. F. Casarin, A. Moretti, An international review of cultural consumption research. SSRN Electron. J. Department of Management, Università Ca' Foscari Venezia working paper 12 (2011)
57. A.M. Gonzalez, L. Bello, The construct "lifestyle" in market segmentation: the behaviour of tourist consumers. Eur. J. Mark. (2002). https://doi.org/10.1108/03090560210412700
58. D.B. Valentine, T.L. Powers, Generation Y values and lifestyle segments. J. Consum. Mark. (2013). https://doi.org/10.1108/JCM-07-2013-0650

59. U.R. Orth et al., Promoting brand benefits: the role of consumer psychographics and lifestyle. J. Consum. Mark. (2004). https://doi.org/10.1108/07363760410525669
60. C.-S. Yu, Construction and validation of an e-lifestyle instrument. Internet Res. (2011). https://doi.org/10.1108/10662241111139282
61. A.M. Thompson, P.F. Kaminski, Psychographic and lifestyle antecedents of service quality expectations: a segmentation approach. J. Serv. Mark. (1993). https://doi.org/10.1108/08876049310047742
62. J.L.M. Tam, S.H.C. Tai, Research note: the psychographic segmentation of the female market in Greater China. Int. Mark. Rev. (1998). https://doi.org/10.1108/02651339810205258
63. T.F. Srihadi, D. Sukandar, A.W. Soehadi, Segmentation of the tourism market for Jakarta: classification of foreign visitors' lifestyle typologies. Tour. Manag. Perspect. **19**, 32–39 (2016). https://doi.org/10.1016/j.tmp.2016.03.005
64. B. Oates, L. Shufeldt, B. Vaught, A psychographic study of the elderly and retail store attributes. J. Consum. Mark. (1996). https://doi.org/10.1108/07363769610152572
65. T.M.M. Verhallen, R.T. Frambach, J. Prabhu, Strategy-based segmentation of industrial markets. Ind. Mark. Manag. **27**(4), 305–313 (1998). https://doi.org/10.1016/S0019-8501(97)00064-3
66. E.J. Cheron, R. McTavish, J. Perrien, Segmentation of bank commercial markets. Int. J. Bank Mark. (1989). https://doi.org/10.1108/EUM0000000001458
67. S.W. Clopton, J.E. Stoddard, D. Dave, Event preferences among arts patrons: implications for market segmentation and arts management. Int. J. Arts Manag. 48–59 (2006)
68. A. Buratto, L. Grosset, B. Viscolani, Advertising a new product in a segmented market. Eur. J. Oper. Res. **175**(2), 1262–1267 (2006)
69. R. Sánchez-Fernández, M. Ángeles Iniesta-Bonillo, A. Cervera-Taulet, Exploring the concept of perceived sustainability at tourist destinations: a market segmentation approach. J. Travel Tour. Mark. **36**(2), 176–190 (2019)
70. K. Bijak, L.C. Thomas, Does segmentation always improve model performance in credit scoring? Expert Syst. Appl. **39**(3), 2433–2442 (2012). https://doi.org/10.1016/j.eswa.2011.08.093
71. A. Sell, P. Walden, Segmentation bases in the mobile services market: attitudes in, demographics out, in *2012 45th Hawaii International Conference on System Sciences* (IEEE, 2012)
72. A. Sell, J. Mezei, P. Walden, An attitude-based latent class segmentation analysis of mobile phone users. Telemat. Inform. **31**(2), 209–219 (2014)
73. D.J. Ketchen, C.L. Shook, The application of cluster analysis in strategic management research: an analysis and critique. Strateg. Manag. J. **17**(6), 441–458 (1996)
74. G. Punj, D.W. Stewart, Cluster analysis in marketing research: review and suggestions for application. J. Mark. Res. **20**(2), 134–148 (1983)